

Samsung T7 Shield External SSD 4TB

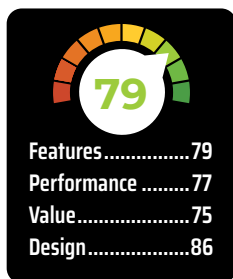
A bit behind the times

The Samsung Portable SSD T7 Shield is the latest addition to the company's rugged line of portable SSDs, and joins the likes of the designed for outdoor enthusiasts and content creators in need of a robust and reliable storage solution. This device is not only built to withstand the elements and the wear and tear of professional life on the move but is also suitable for any user needing to store and transfer large volumes of data quickly and securely. The T7 Shield is a welcome upgrade to the T7 Touch which was launched back in early 2020. While the T7 has been around for over a year, its competitors from WD and Seagate have been around for longer, and have brought newer features to the market much earlier. More on that later. Let's have a look at the build.

BUILD AND DESIGN

If you've ever held the Samsung T7 Touch in your hand, then the Samsung T7 Shield is pretty much the same size. Just that you now have a thick shock-absorbing sheath made of a durable polymer. The ridges along the length of the drive are new and it certainly gives you plenty of places to grip the drive properly. Moreover, Samsung has a little bit of the polymer sheath jutting out from either end. This is pretty good for handling drops wherein the drive falls on the edges. Those are the more painful drops.

₹22,999



The Samsung T7 Shield uses a Type-C port as do most of the portable SSDs with a high-speed interface. The Type-C interface is a USB 3.2 Gen2 10 Gbps port. It is quite dated since most manufacturers already had 20 Gbps drives out by late 2020 such as the Sandisk Extreme Pro V2 external SSD. Then there's even a little white LED indicator beside the port to let you know when the drive is in use. Underneath the polymer sheath, you can actually find an aluminium casing that holds the main PCB inside. So think of the T7 Shield as the T7 with a nice rubbery jacket around it. The entire drive is about 59 mm wide, 88 mm long and 13 mm thick so it is quite slim and easily fits in your pocket. And with a weight of about 98 grams, it doesn't even feel like you're carrying anything. Inside the aluminium casing is a black frame holding the PCB. So the unit is pretty

well built from the looks of it. The PCB has a huge pink thermal pad that sits on one side and spans the entire length and breadth of the PCB. NVMe controllers and even the NAND chips tend to generate quite a lot of heat during sustained write operations and having a decent thermal solution to dissipate all that heat is always welcome and Samsung seems to be doing a pretty good job at it.

All of the above design decisions are what help protect the drive against dust ingress and some protection from water. It's rated IP65 which is also something to boast about. The only downside is that the polymer sheath tends to attract dust and lint, but the upside is that you'll have a cleaner pocket each time you pull it out of your pocket. We jest, of course. The package comes with a USB Type-C to Type-A cable as well as a Type-C to Type-C cable.

THE CIRCUIT

Upon a closer look at the PCB, we can see a couple of familiar faces. Starting with the controller, we have the Samsung S4LR033 which is what Samsung has also used in other drives such as the immensely popular Samsung 980 NVMe SSD. Not the 980 Pro though. The S4LR033 is a DRAM-less controller and that usually means that you take a performance hit. This would have been true a couple of years ago but controllers have come a long way since then especially considering that Samsung makes some of the best controllers. Also, manufacturers have found creative ways of getting around not having a DRAM for a buffer. A portion of the unused space on the SSD is used as a temporary cache by current-gen controllers and that works really well until you fill up about 50-60 per cent of the overall capacity. Beyond that, we might see a degradation